

- 1 (i) Given that $y = \frac{\ln x}{x}$, find $\frac{dy}{dx}$ in terms of x . [2]

(ii) Hence, or otherwise, find the exact value of $\int_1^e \frac{\ln x}{x^2} dx$, showing your working. [4]

$$(i). y = \frac{\ln x}{x} \quad \therefore \frac{dy}{dx} = \frac{x\left(\frac{1}{x}\right) - (\ln x)(1)}{x^2}$$

$$= \frac{1 - \ln x}{x^2} = \frac{1}{x^2} - \frac{\ln x}{x^2}$$

$$(ii). \int_1^e \frac{\ln x}{x^2} dx = -\int_1^e \frac{-\ln x}{x^2} dx = -\int_1^e \frac{1}{x^2} - \frac{\ln x}{x^2} - \frac{1}{x^2} dx$$

$$= -\int_1^e \left(\frac{1}{x^2} - \frac{\ln x}{x^2} \right) - x^{-2} dx$$

$$= - \left[\frac{\ln x}{x} + \frac{1}{x} \right]_1^e$$

$$= - \left[\left(\frac{\ln e}{e} + \frac{1}{e} \right) - \left(\frac{\ln 1}{1} + 1 \right) \right]$$

$$= - \left[\frac{2}{e} - 1 \right]$$

$$= \underline{\underline{1 - \frac{2}{e}}}$$

- 3 (i) It is given that $x \frac{dy}{dx} = 2y - 6$. Using the substitution $y = ux^2$, show that the differential equation can be transformed to $\frac{du}{dx} = f(x)$, where the function $f(x)$ is to be found. [3]

- (ii) Hence, given that $y = 2$ when $x = 1$, solve the differential equation $x \frac{dy}{dx} = 2y - 6$, to find y in terms of x . [4]

(i). given $x \frac{dy}{dx} = 2y - 6$ and $y = ux^2$
 $\therefore x[2ux + x^2 \frac{du}{dx}] = 2ux^2 - 6$ differentiate both sides
 $x^3 \frac{du}{dx} = -6$ w.r.t x :
 $\frac{du}{dx} = \frac{-6}{x^3}$ $\frac{dy}{dx} = u(2x) + x^2 \frac{du}{dx}$

(ii). $\int du = \int -6x^{-3} dx$
 $u = \frac{-6x^{-2}}{-2} + C$
 $u = \frac{3}{x^2} + C$
 $\frac{y}{x^2} = \frac{3}{x^2} + C$
 $\therefore y = 3 + Cx^2$
 Given $y = 2$ when $x = 1$
 hence, $2 = 3 + C$
 $C = -1$
 $\therefore \underline{y = 3 - x^2}$